

b) Show that C and D represents the same linear operator $T: R^2 \rightarrow R^2$ by showing $P^{-1}CP = D$, where $P = P_{B' \rightarrow B} = \{[u'_1]_B \ [u'_2]_B\}$ & $P^{-1} = P_{B \rightarrow B'}$, $B' = \{u'_1, u'_2\}$, $B = \{e_1, e_2\}$, $u'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $u'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Here P & P^{-1} represents the transition matrices.

Question #5[20]: Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$ define the column vectors $u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ and $u_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Use the

Gram-Schmidt process to find the orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then find orthonormal set of vectors $\{q_1, q_2, q_3\}$ by considering standard inner product between the vectors. Also find a matrix R and verify $A = QR$ where

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$$

Question # 6[10]: Suppose $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. Define inner product on A and B as

$$\langle A, B \rangle = \text{tr}(B^T A)$$

where $\text{tr}(C)$ means trace of matrix C .

a) Show that it is a valid inner product by checking its inner product axioms.

b) Also compute $\langle A, B \rangle = \text{tr}(B^T A)$ and $\|A\|$ if $A = \begin{bmatrix} 2 & 5 \\ 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ 6 & 2 \end{bmatrix}$.

Question # 7[20]: Let $T: R^2 \rightarrow R^2$ be defined as $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ -x_1 \\ 0 \end{bmatrix}$. Find the matrix $[T]_{B, B}$

relative to the basis $B = \{u_1, u_2\}$ and $B' = \{v_1, v_2, v_3\}$, where

$$u_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

National University of Computer and Emerging Sciences, Lahore Campus



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Section:	ALL	Page(s):	2
Exam Type:	Final		

Student Name: _____

Roll No. 15-4521 Section: EE-2

Instruction/Notes:

1. Programmable calculator is not allowed.
2. Show complete working in all questions.

Question # 1(a)[7]: Use elementary matrices to find the inverse of $A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$. Also verify that $A = E_1^{-1}E_2^{-1}E_3^{-1} \dots E_{k-1}^{-1}E_k^{-1}$ for some k . Any other method used to evaluate the inverse will not be considered for marking.

b)[3]: Suppose $A = E_1^{-1}E_2^{-1}E_3^{-1} \dots E_{k-1}^{-1}E_k^{-1}$ where A is given in part(a). Consider a unit square with one corner at the origin in the first quadrant of R^2 . Assume the elementary matrices obtained in part (a) as operators and perform the following:

1. Operate E_1^{-1} from left on the unit square and show its effect/image graphically in R^2 .
2. Operate E_{k-1}^{-1} on the image obtained in the above step-1 & show its effect graphically in R^2 .
3. Continue these operations till we reach E_1^{-1} . Show final output graphically in R^2 .

Question # 2[10]: Find vector and parametric equation of the plane in R^3 that passes through the origin and is orthogonal to $v = (3, 1, -6)$.

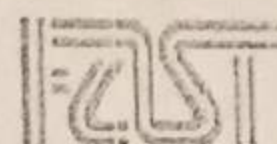
Question # 3[4+2+2+2]: Let $T: R^3 \rightarrow R^3$ be a linear operator $T(x_1, x_2, x_3) = (0x_1 + x_2 - x_3, x_1 + 0x_2 + 2x_3, -1x_1 + x_2 + 0x_3)$ defined by $T(X) = AX$ as, Where $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$.

- a) Check whether T is One to One and Onto.
- b) Find Kernel of T and Basis for Kernel of T .
- c) Find Range of T and Basis for Range of T .
- d) Find Null space of T and Row space of T .

Question # 4[20]: If $C = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then

- a) Find a matrix P (consisting of the Eigen vectors of C) using Eigenvalues of C & show that $P^{-1}CP = D$.

PTO



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Course Title	Linear Algebra(EE)	Course Code	MT104
Midterm — I		Sections: EE(1,2,3)	

Question # 1: Solve the linear system if possible and use the result to determine whether the system has unique, infinitely many or no solution.

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x - 3y + 2z &= 14 \\ 3x + y - z &= -2 \end{aligned}$$

1
2
3

Question # 2: Solve the system of linear equations by using Gauss Jordan elimination method.

$$\begin{aligned} 3x - 2y + 8z &= 9 \\ -2x + 2y + z &= 3 \\ x + 2y - 3z &= 8 \end{aligned}$$

9
3
8

Question # 3: Express the following matrix as product of elementary matrices.

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

Question # 4: Find all values of k for which the given augmented matrix corresponds to a

- (a) Consistent linear system
- (b) Inconsistent linear system

$$\begin{bmatrix} 1 & k & -4 \\ 4 & 8 & 2 \end{bmatrix}$$



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Exam:	Midterm-II	Roll No:	

Instruction/Notes: Programmable calculators are not allowed.

Question # 1 (a) Find the distance between these two parallel planes. (4)
 $x + 2y - 2z = 3$ and $2x + 4y - 4z = 7$

(b)
$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

For the matrix given above find Row space, dimension of row space, Column space, dimension of column space, Rank and Nullity. (6)

Question # 2 Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $B' = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$ are two basis (2+2+2+4)
 for a coordinate vector v in R^2 .

- (a) Find a transition matrix P to change a coordinate vector from B to B'
 (b) Find a transition matrix P' to change a coordinate vector from B' to B
 (c) the vector v is given with coordinates $[v]_{B'} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ relative to basis B'
 Find its coordinates relative to basis B .

Question # 3 (a) Determine Eigen values and Eigen vectors for one of the Eigen value for the following matrix. (6)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

- (b) Let V be the set of polynomials of degree 3 exactly, show that under usual addition and scalar multiplication it is not a vector space, mention which property is not satisfied (4)